Mentimeter

CSCI218: Foundations of Artificial Intelligence

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

CSCI218

Lab 5 (Search)

https://colab.research.google.com/drive/1CdS-G3sNg7rdVjb-N5-UOhe7ICDgBeVS?usp=sharing

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Agenda

Assessed Lab 2 - Search

Recap of Search Algos

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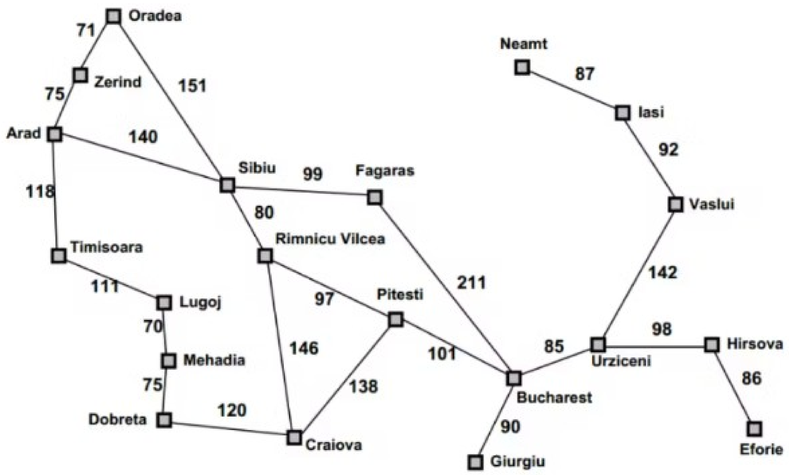
Solving Problems by Searching



Overview

Searching is an important approach used by intelligent agents to solve numerous problems including the travelling salesman problem, the 8-puzzle problem, the N-queens problem, and any practical problems that can be formulated as a searching task in a state space. In this subject, we have studied two types of search methods and their specific algorithms, which are listed as follows.

- Uninformed search: Breadth-first search, Depth-first search, and Uniform-cost search.
- Informed search: Best-first search, Greedy best-first search, and A* search.

Assessed Lab 2 - Search

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Recap of common search algorithms

DFS

BFS

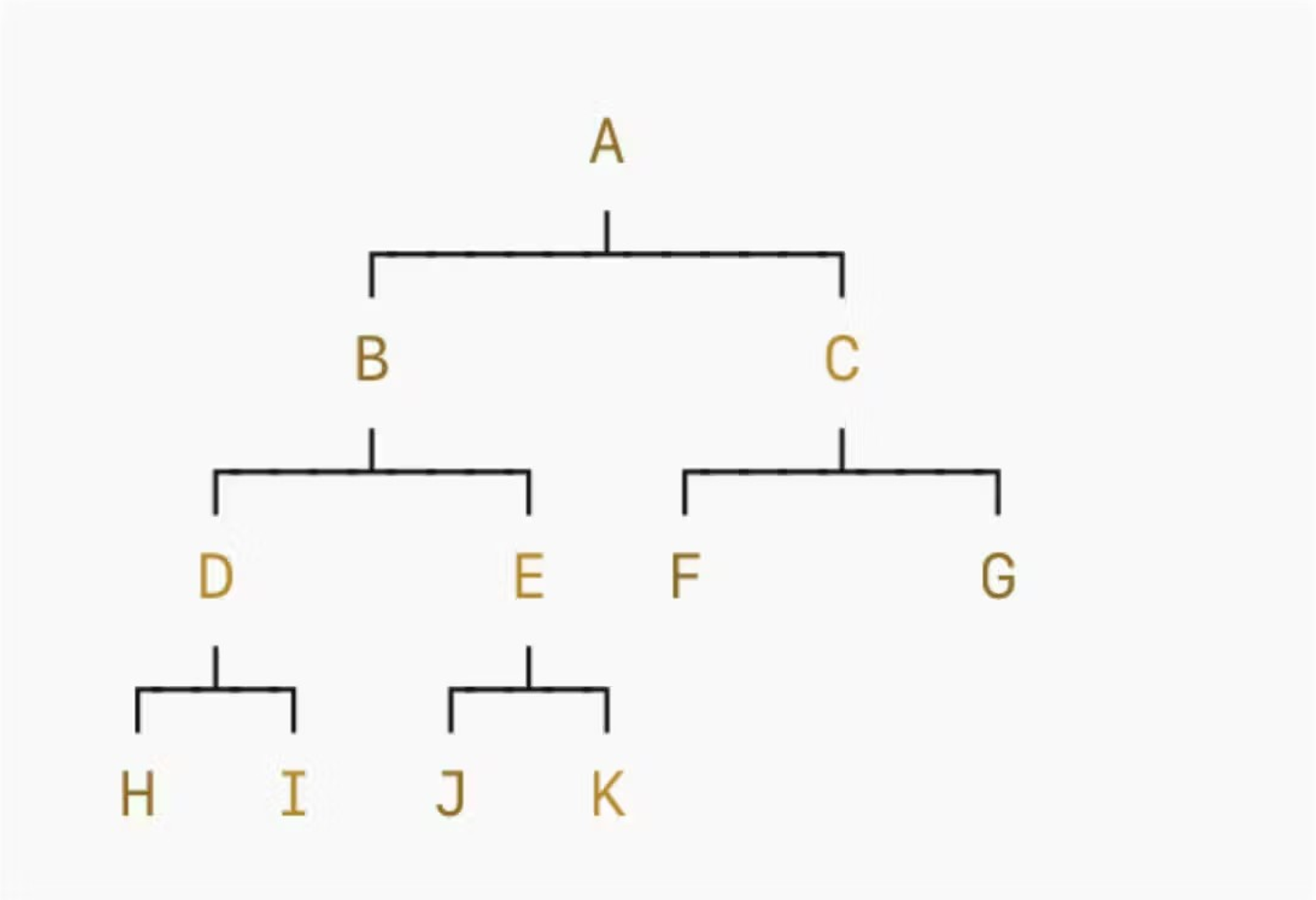
UCS

A* Search

Greedy Best First Search

Search Tree

- Start = A
- Goal state = K
- Children generated left to right
- No early stopping until goal state is reached.

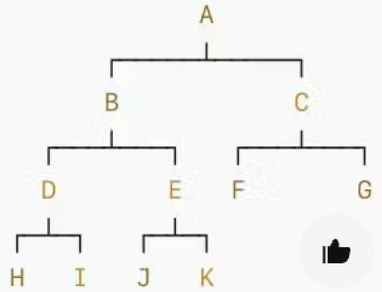


DFS

Step	Expanded	Parent	Frontier: Stack (LIFO)
0	0		A
1	A		C, B
2	B	A	C, E, D
3	D	B	C, E, I , H
4	H	D	C, E, I
5	I	D	C, E
6	E	B	C, K, J
7	J	E	C, K
8	**K**	E	C

DFS path found via parents: A → B → E → K

In DFS, the frontier is a stack, the most recent item is pushed onto the stack first. Therefore, the most recently added node is expanded next.

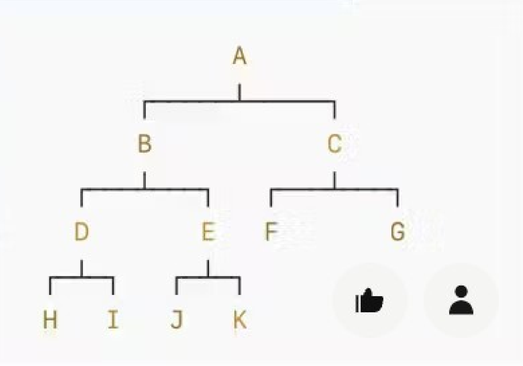


BFS

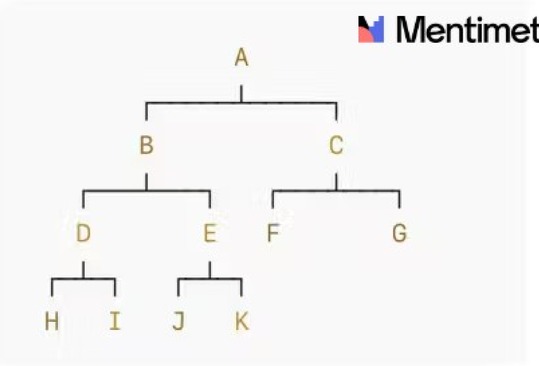
BFS path found via parents: A → B → E → K

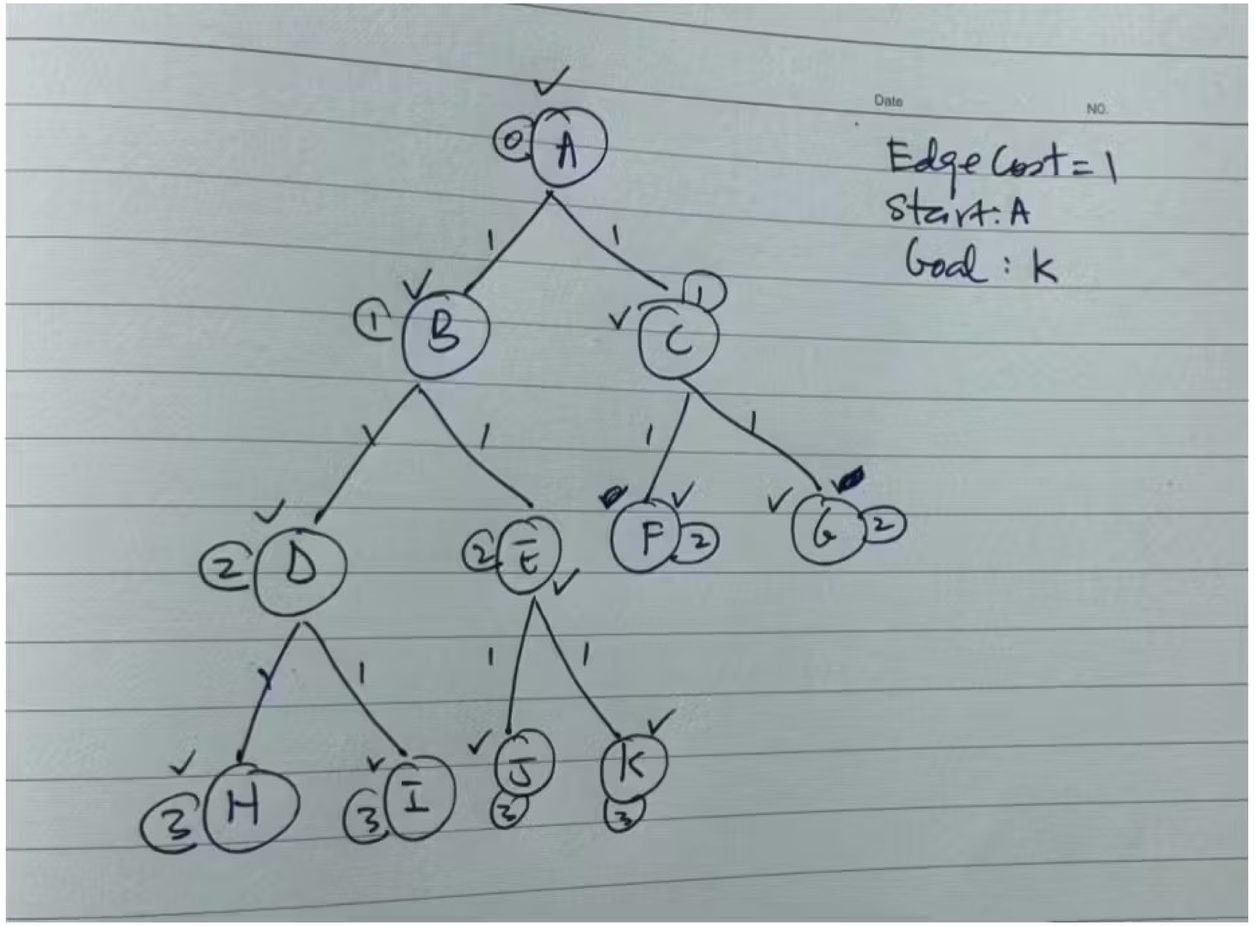
In BFS, the frontier is a queue, so the earliest discovered node is expanded next.

Step	Expanded	Parent	Frontier: Queue (FIFO)
0	0		A
1	A		B ,C
2	B	A	C, D, E
3	C	A	D, E, F, G
4	D	B	E, F, G, H, I
5	E	B	F, G, H, I, J, K
6	F	C	G, H, I, J, K
7	G	C	H, I, J, K
8	H	D	I, J, K
9	I	D	J, K
10	J	E	K
11	K	E	[ ]



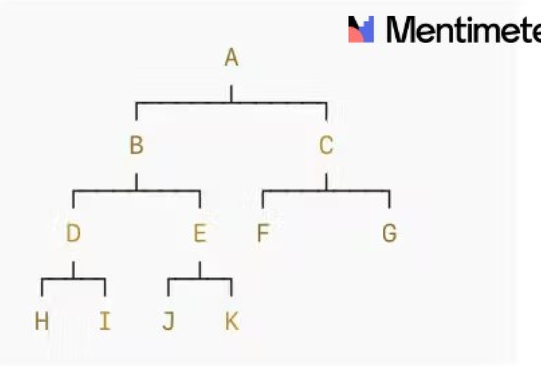
UCS

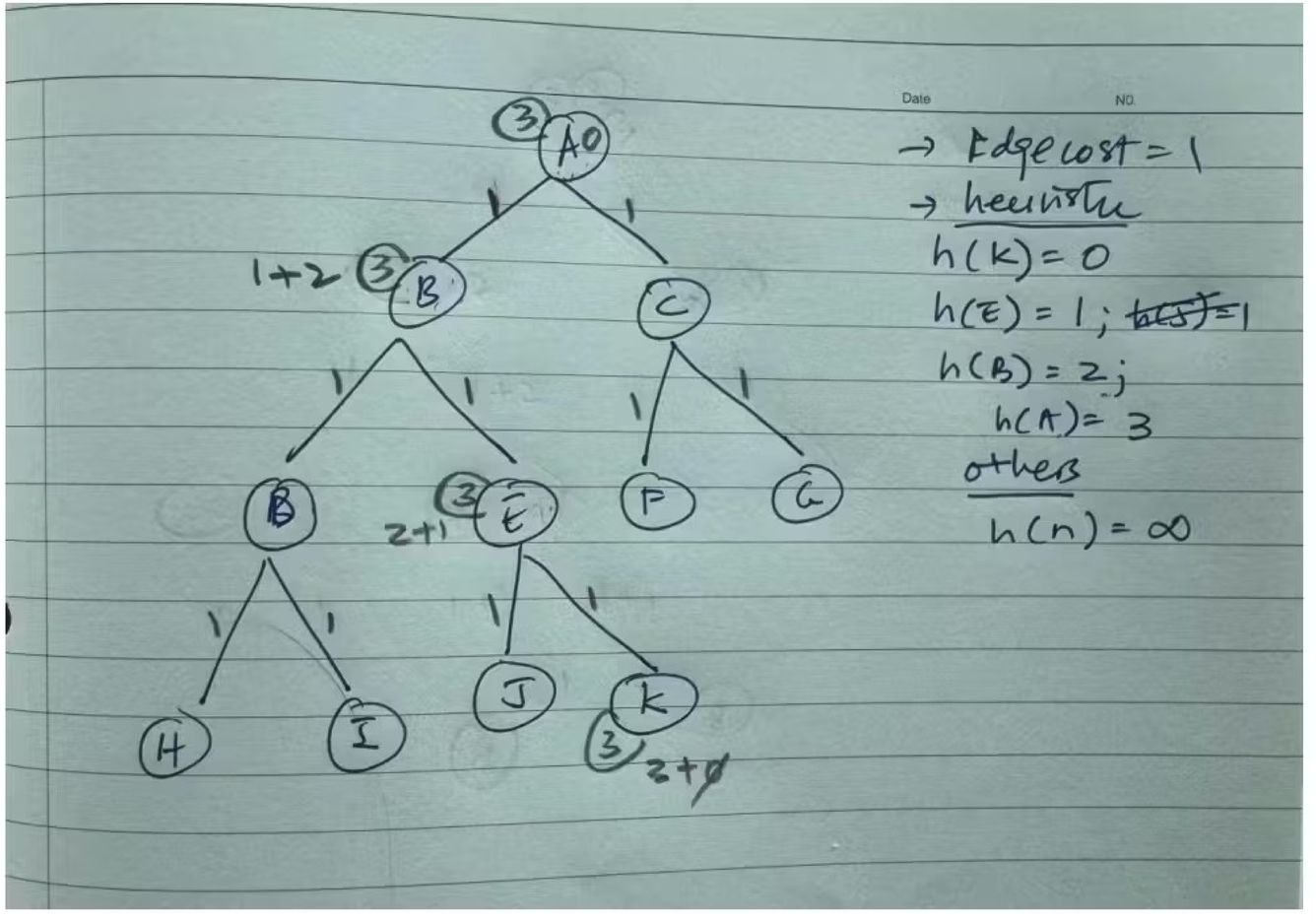




Start: A
Goal: K
Edge cost = 1
Tie break: Left to Right

A* search



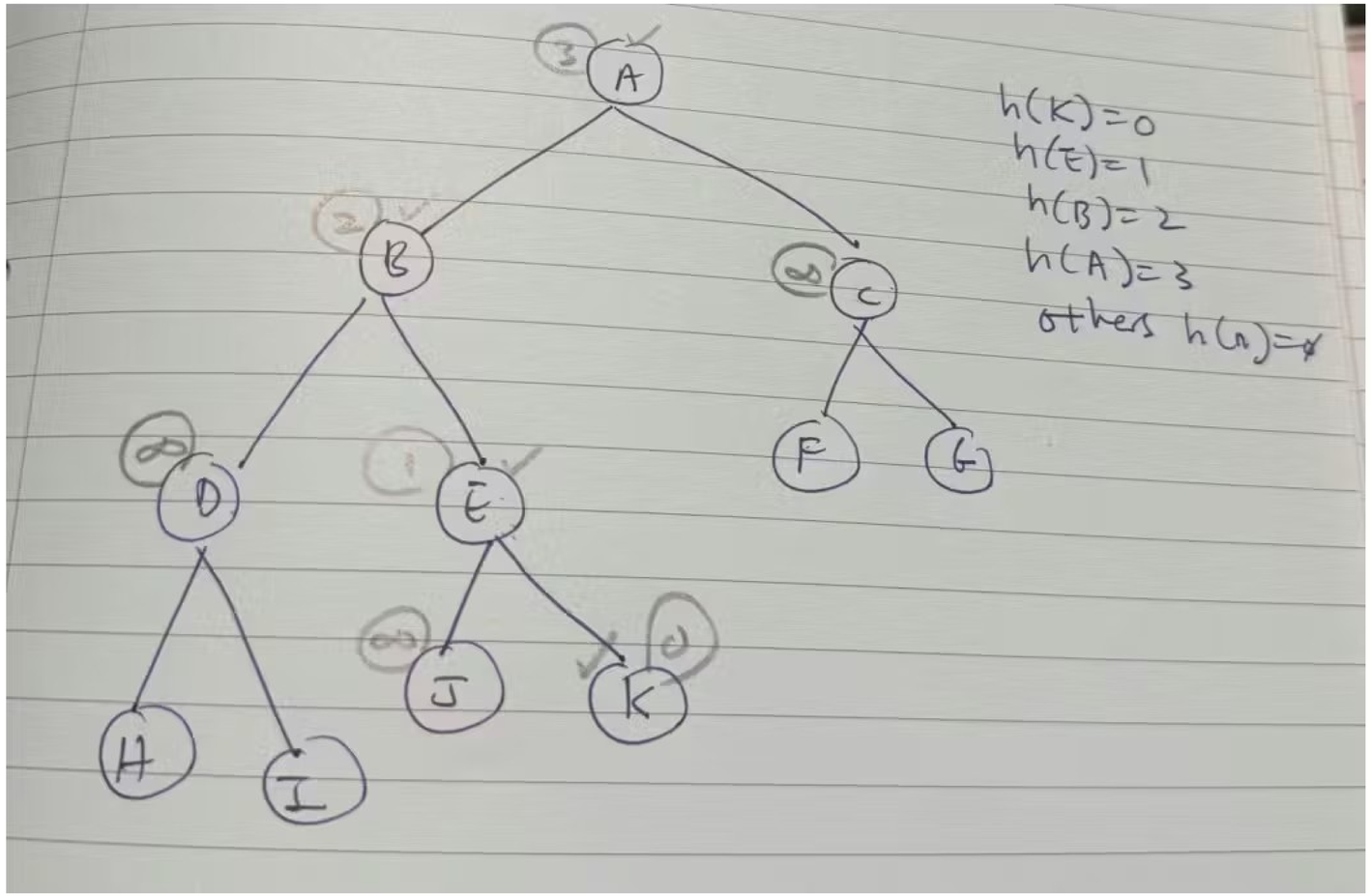


Start: A
Goal: K
Edge cost = 1
Heuristic values to Goal K (distance in edges):

- h(E)=1
- h(B)=2
- h(A)=3
- h(K)=0

Nodes not on K-branch (C,F,G,D,H,I,J) don't lead to K in this tree → treat as ∞ (or "unknown/unreachable").

Greedy Best-First Search



Start: A
Goal: K
Uses h(n) only. Select the node with min h(n).
Heuristic values to Goal K (distance in edges):

- h(E)=1
- h(B)=2
- h(A)=3
- h(K)=0

Nodes not on K-branch (C,F,G,D,H,I,J) don't lead to K in this tree → treat as ∞ (or "unknown/unreachable").

Greedy can work well only if heuristic is perfect. Unlike UCS and A* search which takes into account edge cost, Greedy ignores cost that has already been paid.

Search Comparison

Algorithm	Uses	Guarantee Optimal?
DFS	depth / stack (LIFO)	No
BFS	depth (level order / queue) (FIFO)	Yes (if edge costs equal)
Greedy	h only	No
UCS	g only	Yes
A*	g + h	Yes (if h admissible)